TCP and UDP Protocol

TCP Protocol

The sending and receiving TCP entities exchange data in the form of segments. A TCP segment consists of a fixed 20-byte header (plus an optional part) followed by zero or more data bytes.

TCP Segment Header

Every segment begins with a fixed-format, 20-byte header. The fixed header may be followed by header options.

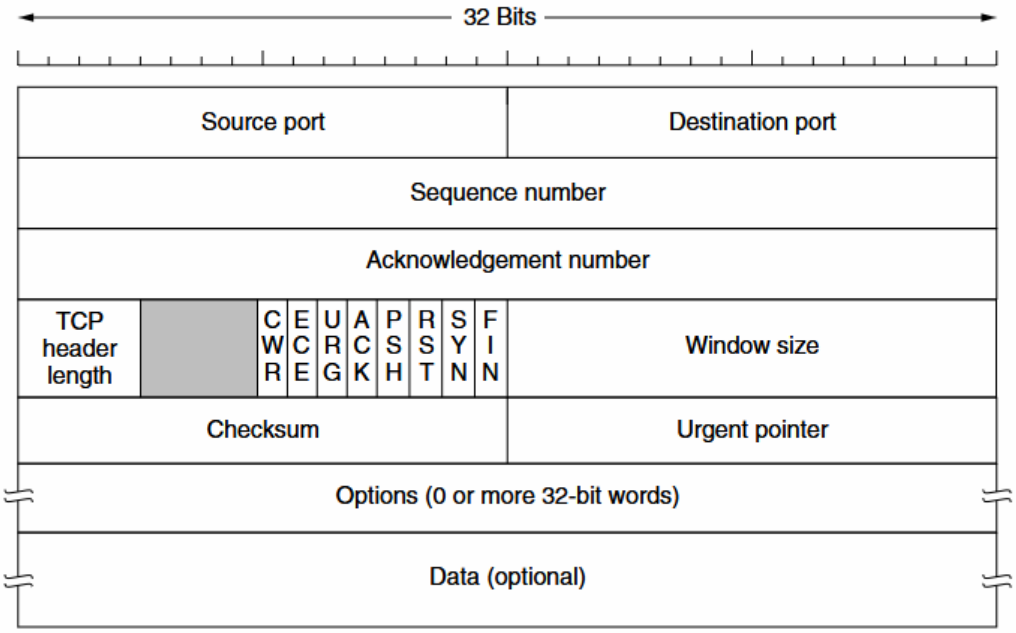


Figure 6-36. The TCP header.

1. **Source port(16 bits) and Destination port(16 bits)** : They are 16 bits in length .The Source port and Destination port fields identify the local end points of the connection. A TCP port plus its host's IP address forms a 48-bit unique end point. The source and destination end points together identify the connection
2. **Sequence number (32 bits) :** The sequence number identifies the position of the first byte in the segment.It helps the receiver arrange data in the correct order. It helps detect missing or duplicate data.

3. **Acknowledgement number (32 bits):** The Acknowledgement Number specifies the sequence number of the next byte expected by the receiver. It acknowledges the successful receipt of data.TCP uses cumulative acknowledgements.It indicates that all bytes before this number have been received correctly.
4. **TCP header length(4 bits) :** The TCP Header Length field indicates the size of the TCP header in bytes.
5. **Flags (CWR | ECE | URG | ACK | PSH | RST | SYN | FIN) :**
   i. CWR : WR is set to signal Congestion Window Reduced from the TCP sender to the TCP receiver so that it knows the sender has slowed down.
   ii. ECE : ECN (Explicit Congestion Notification) .ECE is set to signal an ECN-Echo to a TCP sender to tell it to slow down when the TCP receiver gets a congestion indication from the network.

iii. URG : URG is set to 1 if the Urgent pointer is in use. The Urgent pointer is used to indicate a byte offset from the current sequence number at which urgent data are to be found.

iv. ACK : The ACK bit is set to 1 to indicate that the Acknowledgement number is valid.This is the case for nearly all packets. If ACK is 0, the segment does not contain an acknowledgement, so the Acknowledgement number field is ignored.

v. PSH : The PSH bit indicates PUSHed data. The receiver is hereby kindly requested to deliver the data to the application upon arrival and not buffer it until a full buffer has been received

vi. RST : The RST bit is used to abruptly reset a connection that has become confused due to a host crash or some other reason. It is also used to reject an invalid segment or refuse an attempt to open a connection.

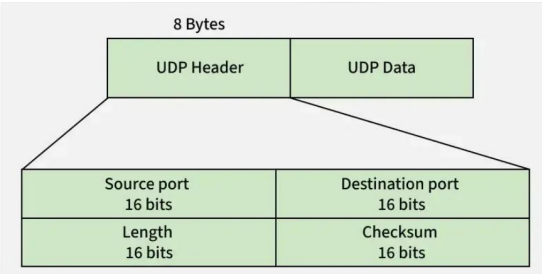
vii. SYN : The SYN bit is used to establish connections. The connection request has SYN = 1 and ACK = 0 to indicate that the piggyback acknowledgement field is not in use. The connection reply does bear an acknowledgement, however, so it has SYN = 1 and ACK = 1. In essence, the SYN bit is used to denote both CONNECTION REQUEST and CONNECTION ACCEPTED, with the ACK bit used to distinguish between those two possibilities.

viii. FIN : The FIN bit is used to release a connection. It specifies that the sender has no more data to transmit. However, after closing a connection, the closing process may continue to receive data indefinitely.

6. **Window Size(16 bits):** The Window size field tells how many bytes may be sent starting at the byte acknowledged
7. **Checksum (16 bits):** Used for error detection. It helps the receiver determine whether the TCP header or data has been corrupted during transmission.
8. **Urgent Pointer:** The Urgent Pointer is a 16-bit field in the TCP header used to indicate the position of urgent data in a TCP segment.
9. **The Options** field provides a way to add extra facilities not covered by the regular header. A widely used option is the one that allows each host to specify the MSS (Maximum Segment Size) it is willing to accept.

UDP (User Datagram Protocol)

UDP is a connectionless transport layer protocol that provides fast communication between applications over a network.Unlike TCP, UDP does not establish a connection before transmitting data and does not guarantee reliable delivery.



UDP has a simple header of 8 bytes (64 bits) consisting of four fields.

Field	Size	Description
Source Port	16 bits	Identifies the sending application.
Destination Port	16 bits	Identifies the receiving application.
Length	16 bits	Specifies the total length of UDP header and data in bytes.
Checksum	16 bits	Detects errors in the header and data.